Positive Square Energy of Every Cyclomatic-Rank-Eleven Cactus

AI-generated manuscript; no human author is claimed

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Abstract

Let G be a connected cactus of cyclomatic rank eleven on n vertices. We prove $s^+(G) > n$. The sharp cactus doubly nonnegative bound leaves exactly $T^{10}Q$ and T^9PP, where $T = C_3$, $P = C_5$, and Q is a cycle. The first family is closed, for every parity of Q, by the all-rank one-distinguished- cycle theorems. For the second, the exact colored audit is $266 = 253 + 13$. Corrected actual-bridge pruning reduces all disconnected structural rows to $T^9P \mid P$ and $P \mid A_9 \mid P$; no blanket leaf lemma is used. The physical $T^9P|P$ certificate closes 50399/50399 classes; the abstract $P|A_9|P$ owner/theorem verifier plus a graph-theoretic connector lift closes 43151/43151 marked classes. The remaining one-cluster universe closes $115512 = 115502 + 10$ physical incidence classes. The executable certificates materialize cyclic vertices, shared cuts, connector paths, router intervals, attachment obligations, and final owner-induced graphs, and rederive packet theorems only after ownership. Frozen SHA-256 gates and hostile mutations make the checks fail closed, including under `python -O`.

1 Definitions, superadditivity, and scope

All graphs are finite, simple, and undirected. A *cactus* is a connected graph every block of which is a bridge or a cycle. Its cyclomatic rank is $|E| - |V| + 1$, equivalently the number of cyclic blocks. If the adjacency eigenvalues of a graph H are $\lambda_1, \ldots, \lambda_h$, put

$$s^+(H) = \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(H) = \sum_{\lambda_i < 0} \lambda_i^2, \qquad \sigma(H) = s^+(H) - |V(H)|.$$

Write $T = C_3$, $P = C_5$, and A_r for a connected shared-cut cluster of r triangles. A cycle C_q is *hostile* when $q \equiv 1 \pmod 4$.

Lemma 1.1 (Induced-partition superadditivity). *If $V(H) = V_1 \dot\cup \cdots \dot\cup V_k$, then*

$$s^+(H) \geq \sum_i s^+(H[V_i]), \qquad \sigma(H) \geq \sum_i \sigma(H[V_i]).$$

Proof. For a real symmetric matrix A,

$$\operatorname{tr}(A_+^2) = \max_{X \succeq 0} \{2 \operatorname{tr}(AX) - \operatorname{tr}(X^2)\}.$$

Indeed, diagonalize A; pinching X to that eigenbasis preserves positive semidefiniteness and does not decrease the objective, and the scalar maximizer is $x = \max(\lambda, 0)$. Let $A = A(H)$ and let B be the block diagonal matrix with blocks $A(H[V_i])$. In the variational formula for A, test $X = B_+$. Because B_+ is block diagonal, off-block entries of A do not contribute, and

$$s^+(H) \geq 2 \operatorname{tr}(AB_+) - \operatorname{tr}(B_+^2) = \operatorname{tr}(B_+^2) = \sum_i s^+(H[V_i]).$$

Subtracting $\sum_i |V_i| = |V(H)|$ proves the second assertion. This is also the superadditivity theorem of Akbari, Kumar, Mohar, and Pragada [1]. $\square$

We use the already proved theorem that *every* connected cactus of cyclomatic rank ten satisfies $s^+ > |V|$ [4, Theorem 6.1], together with the lower-rank and packet bounds proved in that manuscript and its cited predecessors. Thus, when an actual-bridge cut below produces a connected rank-ten territory containing connector remnants or arbitrary attached trees, Theorem 6.1 applies to that territory literally. We do not invoke a stronger rooted-packet version of the rank-ten theorem. In particular, connected ranks two and three have $\sigma \geq 0$, ranks four through ten have $\sigma > 0$, every nonempty triangular cactus has $\sigma > 0$, $\sigma(TP) > 3/4$, $\sigma(PP) > 0$, and the rooted common-cut and packing-one hostile bounds are tree-uniform. Put

$$\delta = \sqrt{5} - 2 < \frac{1}{4}.$$

The certificate ledgers use these exact bounds, not numerical eigenvalue experiments.

The conclusion of this paper is only for connected cyclomatic-rank-eleven cacti. It neither proves the Akbari–Kumar–Mohar–Pragada–Zhang conjecture for all graphs nor claims a universal resolution of it.

2 The sharp DNN frontier

For an edge-containing graph define

$$\kappa(G) = \sup_{M \succeq 0,\ M \geq 0,\ \mathbf{1}^T M \mathbf{1} > 0} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [3, 2] proves, for b bridge blocks and cyclic blocks $C_{\ell_1}, \dots, C_{\ell_r}$,

$$s^-(G) \leq \kappa(G) = b + \sum_i \kappa(C_{\ell_i}), \qquad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}. \end{cases}$$

Set $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ. A rank-eleven cactus has $|E| = n + 10$ and $b + \sum_i \ell_i = n + 10$. Since $s^+ + s^- = 2|E|$, the sharp estimate gives

$$\sigma(G) \geq 10 - \sum_{i=1}^{11} \epsilon_{\ell_i}. \tag{1}$$

Proposition 2.1 (Exact frontier)**.** *The right side of* (1) *is nonpositive only for*

$$T^{10}Q \quad (q \geq 3,\ \textit{including } Q = T), \qquad T^9 PP.$$

Every other rank-eleven cycle multiset has positive surplus.

Proof. The odd sequence ϵ_ℓ decreases strictly, $\epsilon_3 = 1$, and $a := \epsilon_5 = 5 - 2\sqrt{5}$. Exact integer comparisons give

$$0 < a < 1, \quad 3a < 2, \quad 2a > 1, \quad \epsilon_5 + \epsilon_7 < 1.$$

For the first three, after moving radicals to positive sides, the squaring certificates are respectively $20 < 25$, $4 < 5$, $169 < 180$, and $80 < 81$. For the last, $\cos(\pi/7) > 7/8$ gives $\epsilon_7 < 7/15$, while $7/15 < 2\sqrt{5} - 4$ follows from $4489 < 4500$.

With at most eight triangles, the excess sum is at most $8+3a<10$. With exactly nine triangles, two nontriangles add less than one unless both are pentagons; the pair PP adds $2a>1$. With at least ten triangles the multiset is $T^{10}Q$ and its excess is $10+\epsilon_q \geq 10$. These cases are exhaustive. The independent exact arithmetic verifier freezes precisely this classification and remains active under optimization [5]. □

3 Parity-complete closure of $T^{10}Q$

The $T^{10}Q$ residual needs no finite rank-eleven incidence census. Two rank-uniform theorems cover complementary congruence classes.

Proposition 3.1. *Every connected cactus whose cyclic blocks are ten triangles and one cycle $Q = C_q$ satisfies $s^+ > |V|$.*

Proof. If $q \equiv 1 \pmod 4$, the all-rank triangle–hostile-cycle theorem applies to arbitrary shared cuts, bridge connectors, and attached trees [6].

Suppose that q is even or $q \equiv 3 \pmod 4$. Choose a maximum collection of vertex-disjoint cycles, prioritizing a triangle when q is even, and assign each vertex to the lexicographically nearest selected cycle. The resulting induced territories are connected and have cycle-packing number one: otherwise two disjoint cycles in one territory could replace its selected center and enlarge the packing. In a packing-one territory, the grouped Sachs expansion on *it* has

$$\operatorname{Im}\Psi_H(t) = -2\sum_{C\equiv 3\ (4)} Z_{H-V(C)}(t) < 0$$

whenever it contains a triangle; singleton even-cycle terms are real. The continuous spectral argument therefore lies in $(-\pi, 0)$, and the signed Coulson identity gives $s^+(H) > s^-(H)$, hence $s^+(H) > |E(H)| \geq |V(H)|$. A possible territory containing only the even Q is connected, bipartite, and unicyclic, so spectral symmetry gives $s^+ = s^- = |E| = |V|$. At least one prioritized triangle territory is strict. Lemma 1.1 now gives strictness for G. This is the complete nonhostile theorem and proof recorded in [7].

The three cases $q \equiv 1, 3 \pmod 4$ and q even exhaust all q, including $q = 3$. Thus no parity case is omitted. □

4 The exact T^9PP reduction

Join cyclic blocks when they share a cut vertex; the maximal components are *shared-cut clusters*. Contract their full supports in the cactus block–cut tree, take the minimal subtree spanning the cluster nodes, and suppress only unmarked degree-two vertices. The resulting reduced cluster tree has marked leaves. Each reduced edge represents a nonempty chain of actual bridge blocks.

When a reduced edge is cut below, we choose an actual bridge on that chain. Each connector remnant stays with its endpoint component, and each off-hull tree stays with its unique attachment. The two sides are connected, induced, disjoint, and exhaustive. This convention is essential: reduced edges are not contracted analytic edges.

The exact colored partition audit [8] gives

$$267 \text{ total}, \qquad 266 \text{ proper}, \qquad \boxed{266 = 253 + 13},$$

where all 253 direct rows have a positive sum of proved rank-at-most-ten packet bounds. The thirteen structural rows are

$$P|P|9T, \quad P|P|2T|A_7, \quad P|P|T|A_8, \quad P|P|A_9,$$
$$P|7T|T^2P, \; P|6T|T^3P, \; P|5T|T^4P, \; P|4T|T^5P,$$
$$P|3T|T^6P, \; P|2T|T^7P, \; P|T|T^8P, \quad P|T^2P|A_7, \quad P|T^9P.$$

Here kT denotes k singleton clusters, not A_k.

Proposition 4.1 (Corrected actual-bridge pruning)**.** *Every disconnected structural T^9PP configuration has a positive induced actual-bridge partition or reduces to exactly one of*

$$\boxed{T^9P_0 \mid P_1}, \qquad \boxed{P_0 \mid A_9 \mid P_1}.$$

Proof. We use a leaf only when the two displayed packet bounds have positive sum; we do not assert that every strict leaf pays an arbitrary hostile complement.

In $P|P|9T$, remove a singleton-triangle leaf. If none is a leaf, the two pentagons are the only possible leaves, so the reduced tree is a path; a terminal TP and its rank-nine complement are both strict. The same argument applies to $P|sT|T^kP$ for $2 \le k \le 8$ and $s = 9 - k$: absent a singleton leaf, the two nonsingleton marks are the path ends and the bare P groups with its nearest T.

For $P|P|2T|A_7$, a singleton leaf leaves a strict rank-ten complement, while a leaf A_7 leaves a strict rank-four complement. If neither occurs, the pentagons are path ends; the first internal mark from an end gives either $TP+$ rank nine or A_7P+ rank three, with at least one strict summand. For $P|P|T|A_8$, a leaf T or A_8 is certified against respectively rank ten or rank three; otherwise the pentagons are path ends and the first internal mark gives $TP+$ rank nine or A_8P+TP. In $P|P|A_9$, a leaf A_9 gives strict A_9 plus nonnegative PP; otherwise A_9 lies between the two pentagons, which is $P_0|A_9|P_1$.

For $P|T^2P|A_7$, if A_7 is a leaf, its complement has rank four and is strict. Otherwise T^2P is a leaf; it is nonnegative and its rank-eight complement is strict. This includes a three-arm Steiner topology and shows why a blanket leaf statement would be both unnecessary and unsafe. Finally, $P|T^9P$ has two marks and is exactly $T^9P_0|P_1$. Every separation uses the actual-bridge ownership convention above, proving exhaustion [9]. □

5 Physical endpoint certificates

For a shared-cut cluster, its bipartite cycle–cut incidence graph is a tree: an incidence cycle would create a cycle using several blocks or make two blocks meet twice. An external connector is projected to its first hull position, which may be a shared cut or a private cyclic vertex.

A *physical certificate* replaces each abstract incidence by named cyclic positions and edges. Positions carrying one cut are identified; all others remain distinct. Every connector is an ordered path with a root and remnant, and every physical anchor has a separate arbitrary-forest attachment obligation. A router operation partitions a C_3 or C_5 into nonempty proper consecutive intervals. Submitted owner domains must partition an independently reconstructed vertex, edge, and attachment domain. Each owner-induced graph is checked for connectivity, cactus rank, and complete owned cycles. Only then is its theorem and exact symbolic bound selected from a closed whitelist. Unknown profiles, missing owners, split cycles, malformed intervals, or forged bounds raise `RuntimeError`.

The standalone A_9 executable has a narrower integrity boundary. It regenerates the abstract incidence trees and ordered marked classes, realizes every routed triangle by three concrete cyclic positions through the shared interval core, resolves cuts and marks recursively, and rederives terminal theorem records. It represents the two complete pentagon demands abstractly; it does not materialize arbitrary-length external bridge connectors. The next lemma supplies that graph-theoretic lift, so we do not call the standalone ordinary A_9 stream a complete physical connector certificate.

Lemma 5.1 (Abstract-to-physical two-connector lift)**.** *Let A be a shared-cut cluster of nine triangles with its concrete cycle–cut incidence tree and concrete three-position realization of every triangle. Let two complete external pentagon clusters be joined to marked hull positions z_0, z_1 by the two ordered bridge/tree connectors arising in the reduced cluster tree; connector lengths are arbitrary. Suppose an abstract A_9 certificate does all of the following:*

1. *assigns every retained triangle, mark, and pentagon demand to one final owner, and supplies the verifier-derived complete position/cut owner map;*
2. *realizes each sacrificed router triangle by nonempty proper consecutive intervals, consistently under nested refinement; and*
3. *passes the exact final-position gates: every shared cut (including one incident only with sacrificed routers) and every private sacrificed-router position resolves from the root to exactly one terminal owner; every router interval position has the same owner through its child; and the owner records have the independently derived exact domain and no duplicate keys.*

Then the assignment lifts canonically to an induced, disjoint, exhaustive vertex partition of the original $P_0 \mid A_9 \mid P_1$ cactus. Each part is connected, has exactly the complete cyclic blocks in its abstract terminal profile, and has the same cyclomatic rank. The conclusion is uniform in the connector lengths and in arbitrary trees attached anywhere.

Proof. First realize the central cluster. A retained triangle is given wholly to its abstract cycle owner. On every sacrificed triangle, give its three actual vertices to the certified consecutive intervals. At a shared cut all incident retained branches and all router intervals that contain that physical position have the verifier-derived cut owner. The exact gate resolves even a cut incident only to sacrificed routers, rejects multiple child matches, and binds every interval position both from its active child and from the root. Each nonroot active set is the unique child of one parent, so no sibling can retrieve a position from a closed branch. Nested operations refine only the currently active interval. The incidence graph is a tree, so induction from the first router outward shows that the resulting central owner sets are disjoint and exhaustive and that each nonempty set is connected along the incidence branches and interval edges prescribed by the abstract plan.

For $i = 0, 1$, give the complete external P_i, every internal vertex of its ordered connector chain, its connector remnant, and every branch rooted on that chain to the owner of mark z_i. The hull anchor z_i is not added a second time: it is the already owned central vertex at which the chain terminates. If $z_0 = z_1$, there is one physical anchor, not two copies. The exact cut or private-position map and both mark-resolution checks force the two marks and both connector branches to its single owner. The two connector interiors and pentagons remain distinct branches of the cactus block–cut tree, so no other vertex is identified or duplicated. If the marks are distinct, each connector follows its own marked owner. Finally, assign every component off the cyclic hull or a connector to the owner of its unique anchor. Such a component cannot have two anchors, since it and the hull path between them would create another cycle.

These rules assign every original vertex exactly once. Every territory means the subgraph induced by its assigned vertices, so it is induced; edges between different territories are merely discarded by the vertex partition. Connectivity holds centrally by the preceding incidence-tree induction, along each connector because its whole ordered chain and endpoint pentagon have the mark owner, and in every off-hull tree by its unique-anchor assignment.

A retained cyclic block has all its vertices and edges with one owner. A sacrificed router has vertices in at least two proper intervals and therefore is not a cycle in any part. Connectors, remnants, and off-hull components use only bridge blocks and trees. They neither create a cycle nor complete a split router: in a cactus every cycle is already one cyclic block. Consequently the complete cycles in each induced territory are exactly its abstract terminal cycles, and its cyclomatic rank is their number. This also proves independence of connector lengths and attached trees. A detailed standalone formulation is recorded in [11]. □

Proposition 5.2 (The two exact endpoints)**.** *Every graph in* $T^9P \mid P$ *and* $P \mid A_9 \mid P$ *has positive surplus.*

Proof. For $P|A_9|P$, the standalone verifier regenerates 355 unmarked incidence trees, 128155 ordered labelled placements, and 43151 canonical rows. It checks concrete C_3 intervals, every shared cut and sacrificed-router private position, exact duplicate-free final-owner domains, root/child interval owner agreement, unique active-child nesting, mark consistency, and post-ownership theorem records. Its ten hostile mutations include a duplicate or forged final-position map and a shared-cut interval ambiguity. No accepted row has a cut incident only with sacrificed routers, although the exhaustive cut-domain gate does not assume this. Exactly 43145 ordinary plans close. By Lemma 5.1, every one of those plans lifts over both complete external pentagons and their arbitrary ordered connector chains. Fifteen nested two-arm plans have profiles $A_aP + A_bP + A_{7-a-b}$ for $a, b > 0$, $a + b \leq 6$; their weakest ledger is $TP + TP + A_5 > 2 - 2\delta$. The six remaining rows are physically repaired: two split as $TP+$ packing-one A_7P, and four open one named pentagon path at cost -1 while retaining packing-one A_9P. Their pentagon opening geometry is checked explicitly. In those six repairs, the external connector follows its checked connector-root owner and all of its internal bridge vertices follow the same owner; the complementary opened four-vertex pentagon path remains the certified tree territory. The unique-anchor argument in Lemma 5.1 proves that arbitrary connector length and attached trees do not change that repair. Thus $43151 = 43145 + 6$ [10].

For $T^9P|P$, both pentagons, every triangle, connector, remnant, and attachment are physical. There are 8011 colored T^9P incidence trees and 50399 marked rows: 43151 triangular-hull rows and 7248 private clustered-pentagon rows. The first set projects bijectively to the hardened A_9 certificate, closing 43145 ordinary rows and the same six repairs. Of the private rows, 7246 use the first actual triangle beyond the clustered pentagon: its singleton root interval owns both pentagons and a connected child of rank at most nine, while its two-vertex interval owns a nonempty strict triangular packet. The two bouquet-distance orbits open a specified pentagon vertex and retain packing-one A_9P, with ledger $(9 - \delta) - 1 = 8 - \delta > 0$. Hence all 50399 rows close [12].

The $T^9P|P$ verifier is independent corroboration of the local lift for the 43145 ordinary A_9 plans: its exact projection bijection rebinds those plans to concrete triangle positions and then checks complete pentagons, a complete remote connector, remnants, attachments, connectivity, cycles, and ranks. The projection uses ordered labelled A_9 marks. Either orientation of the two external roles is represented after exchanging the labels P_0, P_1 (the ordered placement stream contains the corresponding exchanged marks), so no asymmetry of the abstract plan is hidden. This corroborating executable does *not* by itself enumerate two independently arbitrary nonzero external connector lengths: that uniform extension is the content of Lemma 5.1, whose proof uses only that each added connector is a bridge/tree branch at its marked anchor. □

6 One fully shared T^9PP cluster

If all eleven cycles lie in one shared-cut cluster, the exact color-preserving incidence census is

shared cuts	1	2	3	4	5	6	7	8	9	10	total
all	1	22	264	1790	7560	20080	33154	32369	16775	3497	115512
ordinary	0	20	260	1788	7559	20080	33154	32369	16775	3497	115502
residual	1	2	4	2	1	0	0	0	0	0	10

The ordinary verifier does not trust a single abstract witness. It physically checks all 517923 safe cycle choices over the 115502 rows. In particular, a sacrificed pentagon may have two through five occupied ports; cyclic intervals are checked directly, including the forced $1+1+1+2$ and five-singleton cases. The ten residuals have the following physically rederived closures:

	operation	strict ledger
U1	common-cut T^9PP	$10-4/(3\sqrt{13})$
U2	open P; common-cut T^9P	$8-\delta$
U3	$P+$ common-cut T^8P	$8-2\delta$
U4	C_5 interval: A_8+TP	$3/4$
U5	open P; packing-one T^9P	$8-\delta$
U6	$P+T+$ common-cut T^7P	$7-2\delta$
U7	corrected $P+$ packing-one T^8P	$8-2\delta$
U8	degree-four C_5: $T+$ rank-nine T^8P	>0
U9	nested $P+P+T+A_6$	$1-2\delta$
U10	nested $P+P+T+T+A_5$	$2-2\delta$

In U9 and U10, later routing refines only the active child; a closed sibling cannot be recovered. U7 is the corrected repair, not the false inherited $P+P+A_7$ ledger. The physical verifier therefore proves

$$\boxed{115512 = 115502 + 10 \quad \text{closed classes.}} \tag{2}$$

This is a finite one-cluster theorem, not an all-rank T^rPP assertion [13].

7 Main theorem

Theorem 7.1. *Every connected cactus G of cyclomatic rank eleven, on n vertices, satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. Proposition 2.1 proves the claim unless the cycle multiset is $T^{10}Q$ or T^9PP. Proposition 3.1 closes the first family for all parities of Q.

For T^9PP, partition its cyclic blocks by shared-cut cluster. If there is more than one cluster, the exact proper colored audit has 253 direct positive rows and thirteen structural rows. Proposition 4.1 closes every structural topology by certified actual-bridge cuts or reduces it to exactly $T^9P|P$ or $P|A_9|P$; Proposition 5.2 closes both endpoints. If there is one cluster, the complete physical certificate (2) closes it. In every analytic partition the territories are induced, disjoint, and exhaustive, so Lemma 1.1 adds their surplus bounds. Therefore $\sigma(G) > 0$, which is the assertion. □

8 Certificate integrity, exclusions, and reproduction

All authoritative rank-eleven programs use Python 3.10 or newer, exact integer, `Fraction`, or symbolic radical comparisons, and explicit `require`/`RuntimeError` gates. They are run normally and under `-O`; theorem acceptance does not depend on `assert`. The endpoint and fully shared verifiers reconstruct expected domains independently of submitted certificates, require exact owner-domain equality, rederive terminal packet records after ownership, and reject hostile mutations. The $P|A_9|P$, $T^9P|P$, and fully shared suites reject respectively 10, 31, and 25 mutations.

The principal frozen SHA-256 values are:

certificate	digest
$P|A_9|P$ canonical rows	`0bf53914ae760002386b4b94e4de2d0cccbe61725063b4a46435bcd49c70403b`
$P|A_9|P$ complete owner/theorem records	`58f9951b620fa9f4830724a8bbc5b426a6125437b11c84b125d4ee63488dd3ec`
$P|A_9|P$ repairs	`9b8631b8d1b92970584156e2e444fedf78c2394e0867d43c1204aa09c4f49e0e`
$T^9P|P$ combined geometry	`3da4ebec400a236a10ffb242603b485c7b549a2503fa5c4ee061dcc7afa70b7b`
$T^9P|P$ ordinary physical owners	`63305ff27b19d07bd705eec8f489dcfcfd12cc8cc129dbe93cf914d1c29c4a1a`
$T^9P|P$ private certificates	`815040d4da58efb5edb5660de47d14d4012eb6245afcabb5b77c98e2a8a8e43d`
$T^9P|P$ six repairs	`740a1385503bdf58761be38057ca9d548f85289183ef4a4c515fbc6038398da3`
fully shared canonical rows	`65f4d845ff0ef17ce7880992810de149fd2108927e2ef03b8fac57032ac72ce2`
fully shared ordinary proofs	`5d134b875d7ff369c74f361f4fd58a2ee7262c8bfdaba0453987f46f3391b70e`
all 517923 safe choices	`071df2e10153eb21a8153cc3e45de6768e350a2257a692b0b03979227bc37a0f`
fully shared repairs	`eedc3bebd64e4711849115b3846db2eee2a93cd7ffee628e49f7a3133f73c324`

From the repository root run:

```
python3 research/rank-eleven-cactus-dnn-residual-frontier-verifier.py
python3 -O research/rank-eleven-cactus-dnn-residual-frontier-verifier.py
python3 research/rank-eleven-residual-partition-audit.py
python3 -O research/rank-eleven-residual-partition-audit.py
python3 research/rank-eleven-a9-two-interface-verifier.py
python3 -O research/rank-eleven-a9-two-interface-verifier.py
python3 research/rank-eleven-t9p-p-endpoint-frontier-verifier.py
python3 -O research/rank-eleven-t9p-p-endpoint-frontier-verifier.py
python3 research/rank-eleven-t9pp-fully-shared-first-phase-verifier.py
python3 -O research/rank-eleven-t9pp-fully-shared-first-phase-verifier.py
bash scripts/build-paper.sh all-rank-eleven-cacti
```

The proof deliberately excludes provisional or withdrawn artifacts: the conditional rank-eleven frontier census is not an endpoint theorem; the former $43151 = 43116 + 35$ score is not used; the withdrawn symbolic 50382/50399 closure and its 17-row blueprint are not certificates; and historical router-reachability assertion R11 is neither assumed nor proved. R11 is unneeded because corrected actual-bridge pruning yields only the two exact endpoints above, while the one-cluster case has its own complete physical certificate. No open two-pivot winding statement is used.

AI disclosure

This manuscript was generated and assembled by an AI coding and mathematical reasoning system from the cited repository proofs and executable certificates. No human author, human review, or human verification is represented by the blank-free author line. The claims should be assessed from the displayed arguments, cited inputs, frozen artifacts, and reproducible fail-closed checks, not from an asserted human authority.

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